

attempt to reach the corresponding Node in order to apply these updates. Examples of configuration that may be in such a state include:

1. Node GroupKey additions and removals.
2. Node Endpoint Group membership additions and removals.
3. Node ACL additions, updates and removals for groups, including changes relating to CAT version updates.
4. Node Binding additions and removals for groups.

See [Section 11.24.7.11, “RefreshNode Command”](#) command for a detailed description of each step. An Administrator MAY invoke this command at any time to request that the Datastore perform this operation immediately for a given Node.

When a new node entry is left in the pending state, a [Joint Fabric](#) Administrator may not be able to determine whether commissioning completed or failed. In this scenario, the [Joint Fabric](#) Administrator must continue to monitor and attempt maintenance duties upon the node until the node entry is removed by another [Joint Fabric](#) Administrator or by the user.

11.24.4.4. Adding a Node to a Group

When adding a Node to a Group, if the GroupKeySetID is set, then the [Joint Fabric](#) Administrator SHALL use the [AddGroupIDToEndpointForNode](#) command to update the Node with the KeySet and Group relationship required for the new Group membership.

When the Group membership is intended for control of the Node, the [Joint Fabric](#) Administrator SHALL use the [AddACLToNode](#) command to update the Node with the appropriate ACL.

When the Group membership is intended for control by the Node, the [Joint Fabric](#) Administrator SHALL use the [AddBindingToEndpointForNode](#) command to update the Node with the appropriate Binding.

11.24.4.5. Removing a Node from a Group

When the Group membership was used for control of the Node, the [Joint Fabric](#) Administrator SHALL use the [RemoveACLFromNode](#) command to remove the ACL from the Node.

When the Group membership was used for control by the Node, the [Joint Fabric](#) Administrator SHALL use the [RemoveBindingFromEndpointForNode](#) command to remove the Binding from the Node.

When removing a Node from a Group, if the GroupKeySetID is set, then the [Joint Fabric](#) Administrator SHALL use the [RemoveGroupIDFromEndpointForNode](#) command to ensure the KeySet and Group relationship is removed from the Node and the Datastore reflects the new Group membership.

For groups with access control considerations, the [Joint Fabric](#) Administrator SHALL use the [UpdateGroup](#) command to update the GroupKeySetID (if set) and increment the GroupCATVersion. If the Node is a Controller, then the [Joint Fabric](#) Administrator SHALL also update the GroupCATVersion of the Group.